

Airline Flight Analysis Report

1. Introduction

This report presents a detailed analysis of domestic airline flight data across six major Indian cities. The goal is to understand booking patterns, pricing behaviour, and revenue distribution across airlines, and present these findings through an interactive Excel dashboard.

Data Source: Kaggle

Tool Used: Microsoft Excel

Records Analysed: 300,153 flight bookings

2. Dataset Description

The dataset contains the following fields:

- Airline - Operating the flight (6 airlines)
- Flight - Flight Number
- Source_city - Departure city
- Destination_city - Arrival City
- Departure_time - Time of the day flight departs
- Arrival_time - Time of the day flight arrives
- Stops - Number of stops (Zero, One)
- Class - Travel Class (Economy, Business)
- Duration - Flight Duration in Hours
- Days_left - Number of days left before departure
- Price – Ticket Price

3. Data Cleaning & Preparation

The dataset was cleaned and prepared in Microsoft Excel before analysis to ensure data quality.

The following steps were carried out:

- Checked the dataset for missing values and duplicates records.
- Standardized the formatting of dates, text and numeric values for consistency.
- Corrected inconsistent text entries by removing extra spaces.
- Created a new column “Flight Type” using the IF formula to classify flights as Direct Flights or Connecting Flights based on the number of stops.

4. Key Performance Indicators (KPIs)

- Total Flights - 300,153
- Total Revenue - ₹6.27B
- Average Ticket Price - ₹20.89K
- Most Booked Airline - Vistara
- Average Flight Duration - 12.22

5. Exploratory Data Analysis (EDA)

5.1 Bookings and Revenue by Airline

Airline	Flight Booked	Revenue	Avg. Ticket
Vistara	127,859	3,886,470,735	30,396.54
Air India	80,892	1,901,529,790	23,507.02
Indigo	43,120	229,580,207	5,324.22
GO FIRST	23,173	130,973,972	5,652.01
AirAsia	16,098	65,858,089	4,091.07
SpiceJet	9,011	55,681,482	6,179.28

Vistara recorded the highest number of bookings (127,859) and generated the highest revenue (₹3.89B), making it the top performing airline. Air India ranked second with 80,892 bookings and ₹1.90B in revenue.

Overall, the analysis shows that Vistara was the dominant airline in terms of both passenger demand and revenue generation.

5.2 Class Distribution

Class	Bookings	Total Bookings (%)	Revenue share
Economy	206,666	69	21.7%
Business	93,487	31	78.3%

Although business class accounted for only 31% bookings, it generated 78.3% of the total revenue, highlighting its higher revenue contribution per booking.

5.3 Direct vs Connecting Flights

Flight Type	Flights	Share
Direct	36,004	12%
Connecting	264,149	88%

Connecting flights dominated the dataset, accounting for 88% of all flight records, while direct flights represented 12%.

6. Dashboard Design

The dashboard was designed to present the most important insights from the airline dataset in a clear and interactive format. It includes:

- 5 KPI cards displaying Total Flights, Total Revenue, Average Ticket Price, Most Booked Airline and Average Flight Duration.
- 6 interactive charts showing Passengers' Preferred Airline, Business vs Economy class distribution, Revenue by Airline, Preferred Flight Time, Direct vs Connecting Flights, Revenue of Airline by Class
- Slicers (Airline, Source City, Class and Stops) that allows users to interactively filter the dashboard.

7. Key Insights Summary

- Vistara was the top performing airline accounting for 42.6% of the total bookings and 62% of total revenue making it the leading airline.
- Business Class generated a higher share of the revenue despite fewer bookings. This highlights the higher value of premium fares.
- Connecting flights accounted for majority of bookings, making up 88% of all flight records, while direct flights represented just 12%.
- Morning departures were the most preferred, with Morning and Early Morning flights together accounting for nearly half of all bookings.
- Travel class had a greater impact on ticket prices than booking timing.

8. Limitations

- The days left is limited to a maximum of 49 days, making it difficult to analyze pricing patterns for bookings made further in advance.
- The analysis is based on historical data rather than data collected over multiple time periods. Therefore, seasonal trends and long-term changes in passenger demand could not be assessed.

9. Recommendation

- Enhance the experience of passengers on connecting flights, as they account for most bookings.
- Review the performance of low performing airline top identify opportunities for growth.
- Optimize flight schedules during peak morning hours to meet passenger demand and improve operational efficiency.
- Prioritize resources on high performing routes where demand and revenue are highest.